



LINEAR ALGEBRA

UE19MA251

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Agenda



- Proofs of SVD (Alternate proof)
- Matrices and SVD
- Singular vectors and fundamental subspaces

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Alternate proof of SVD



We show $U^T A V = \Sigma$ which leads to SVD $A = U \Sigma V^T$. For $j = 1 \dots r$, we have $u_j = \frac{A v_j}{\sigma_j}$, so $A v_j = \sigma_j u_j$ and

$$(U^T A V)_{ij} = u_i^T A v_j = u_i^T (\sigma_j u_j) = \sigma_j \delta_{ij}, \quad i = 1, \dots, m$$

For $j = r + 1 \dots n$, we have $(A^T A) v_j = 0$, so $A v_j = 0$ and

$$(U^T A V)_{ij} = u_i^T A v_j = 0, \quad i = 1, \dots, m$$

Combining the results, we get

$$U^T A V = \Sigma$$

Hence

$$A = U \Sigma V^T$$

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Matrices and SVD

For a matrix of order $m \times n$ with SVD $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, the column vectors of $\mathbf{U}$ (resp. $\mathbf{V}$) is an orthonormal basis of $\mathbb{R}^m$ (resp. $\mathbb{R}^n$).

$\mathbf{U}$ must be an eigenvector matrix of $\mathbf{A}\mathbf{A}^T$.

$$\mathbf{A}\mathbf{A}^T = (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T) (\mathbf{V}\mathbf{\Sigma}^T\mathbf{U}^T) = \mathbf{U} \overbrace{(\mathbf{\Sigma}\mathbf{\Sigma}^T)}^{\text{diagonal}} \mathbf{U}^T$$

Similarly, $\mathbf{V}$ must be an eigenvector matrix of $\mathbf{A}^T\mathbf{A}$.

$$\mathbf{A}^T\mathbf{A} = (\mathbf{V}\mathbf{\Sigma}^T\mathbf{U}^T) (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T) = \mathbf{V} \overbrace{(\mathbf{\Sigma}^T\mathbf{\Sigma})}^{\text{diagonal}} \mathbf{V}^T$$

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Singular Vectors and Fundamental Subspaces

The right (resp. left) singular vectors in an SVD of a matrix form an orthonormal basis of the row space (resp. column space) of the matrix.

Row space. $\{v_{r+1}, \dots, v_n\}$ contains eigenvectors of $(A^T A)$ with eigenvalue 0, so it is a basis of $\mathcal{N}(A^T A) = \mathcal{N}(A)$. This implies $\{v_1, \dots, v_r\}$ is a basis of the orthogonal complement of $\mathcal{N}(A)$, i.e. the row space of A .

Column space. We have $AV = U\Sigma$. The first r columns are

$$Av_i = \sigma_i u_i, \quad i = 1, \dots, r$$

So $\{u_1, \dots, u_r\}$ is a linearly independent set in the column space of A . Hence, it is a basis of $\mathcal{C}(A)$.



THANK YOU

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